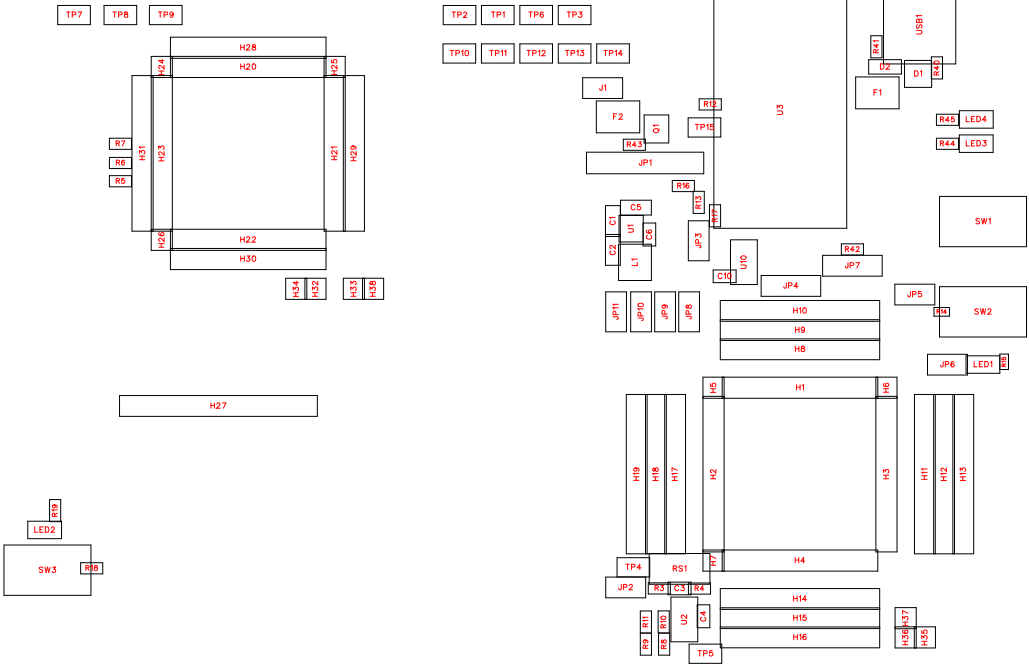

TP7
TP8
TP9
H28
H24
H20
H25
R7
R6
R5
H31
H23
H21
H29
H26
H22
H30
H34
H32
H33
H36
H27
R9
LED2
SW3
R8
TP2
TP1
TP6
TP3
TP10
TP11
TP12
TP13
TP14
J1
F2
Q1
R43
JP1
R12
TP15
U3
USB1
R41
U2
D1
R40
F1
R45
LED4
R44
LED3
R16
R15
R14
C5
C1
U1
C6
C2
L1
JP3
SW1
U10
C10
R42
JP7
JP4
JP5
R4
SW2
JP11
JP10
JP9
JP8
H10
H9
H8
JP6
LED1
R39
H5
H1
H6
H19
H18
H17
H2
H3
H11
H12
H13
H7
H4
TP4
RS1
C3
R4
JP2
R3
H14
H15
H16
R11
R10
C4
U2
R9
R8
TP5
H37
H38
H35

No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier	LCSC Part Name
1	1	10uF	C1	C0805	10uF	CL21A106KAYNNNE	SAMSUNG(三星)	CL21A106KAYNNNE.1	LCSC	
2	2	22uF	C2,C5	C0805	22uF	CL21A226MAQNNNE	SAMSUNG(三星)	CL21A226MAQNNNE.1	LCSC	
3	2	100nF	C3,C4	C0603	100nF	CL10B104KB8NNNC	SAMSUNG(三星)	C1591	LCSC	100nF ±10% 50V
4	1	100nF	C6	C0603	100nF	CL10B104KB8NNNC	SAMSUNG(三星)	CL10B104KB8NNNC.1	LCSC	
5	1	100nF MUX DECOUPLING	C10	C0603	100nF	CL10B104KB8NNNC	SAMSUNG(三星)	CL10B104KB8NNNC.1	LCSC	
6	1	USB ESD	D1	SOT-23-6_L2.9-W1.6-P0.95-LS2.8-BL		USBLC6-2SC6		USBLC6-2SC6_C323793.1	LCSC	
7	1	USB-C VBUS TVS	D2	SOD-123_L2.8-W1.8-LS3.7-RD		SMF5.0A		SMF5.0A.1	LCSC	
8	1	USB-C VBUS PTC 1.1A	F1	F1812		MF-MSMF110/16-2	BOURNS	C210834	LCSC	16V 1.1A 1812 Resettable PTC
9	1	MF-MSMF050-2	F2	F1812		MF-MSMF050-2	BOURNS	C17313	LCSC	15V 500mA 1812自恢复
10	1	DEV_CORE_NORTH	H1	HDR-TH_8P-P2.54-V-M				Header-Female-2.54_1x8.1		
11	1	DEV_CORE_WEST	H2	HDR-TH_8P-P2.54-V-M				Header-Female-2.54_1x8.1		
12	1	DEV_CORE_EAST	H3	HDR-TH_8P-P2.54-V-M				Header-Female-2.54_1x8.1		
13	1	DEV_CORE_SOUTH	H4	HDR-TH_8P-P2.54-V-M				Header-Female-2.54_1x8.1		
14	1	DEV_KNW_SOCKET_TBD	H5	HDR-TH_1P-P2.54-V-M				HDR_1x1.1		
15	1	DEV_KNE_SOCKET_TBD	H6	HDR-TH_1P-P2.54-V-M				HDR_1x1.1		
16	1	DEV_KSW_SOCKET_TBD	H7	HDR-TH_1P-P2.54-V-M				HDR_1x1.1		
17	1	DEV_BREAKOUT_VDD_TARGET	H35	HDR-TH_1P-P2.54-V-M		2.54-1*1P针	BOOMELE(博穆精密)	Header-Male-2.54_1x1.1	LCSC	
18	1	DEV_BREAKOUT_GND1	H36	HDR-TH_1P-P2.54-V-M		2.54-1*1P针	BOOMELE(博穆精密)	Header-Male-2.54_1x1.1	LCSC	

19	1	DEV_BREAKOUT_GND2	H37	HDR-TH_1P-P2.54-V-M		2.54-1*1P针	BOOMELE(博穆精密)	Header-Male-2.54_1x1.1	LCSC	
20	2	PZ254V-11-02P	J1,JP2	HDR-TH_2P-P2.54-V-M		PZ254V-11-02P	XFCN(兴飞)	C492401	LCSC	1x2P 间距:2.54mm 方针 直插
21	1	PWR SOURCE: 1-2 USB-C / 3-4 STLINK / 5-6 EXT; FIT ONE ONLY	JP1	HDR-TH_6P-P2.54-V-M		PZ254V-11-06P	XFCN(兴飞)	PZ254V-11-06P.1	LCSC	
22	1	PZ254V-11-02P	JP3	HDR-TH_2P-P2.54-V-M		PZ254V-11-02P	XFCN(兴飞)	PZ254V-11-02P.1	LCSC	
23	1	BOOT_SEL	JP4	HDR-TH_3P-P2.54-V-M_PZ254V-11-03P		PZ254V-11-03P	XFCN(兴飞)	PZ254V-11-03P_C2937625.1	LCSC	
24	1	USER_BTN_EN (DEFAULT SHORT)	JP5	HDR-TH_2P-P2.54-V-M		PZ254V-11-02P	XFCN(兴飞)	PZ254V-11-02P.1	LCSC	
25	1	USER_LED_EN (DEFAULT SHORT)	JP6	HDR-TH_2P-P2.54-V-M		PZ254V-11-02P	XFCN(兴飞)	PZ254V-11-02P.1	LCSC	
26	1	DBG MODE: 1-2 SWD / 2-3 USB	JP7	HDR-TH_3P-P2.54-V-M_PZ254V-11-03P		PZ254V-11-03P	XFCN(兴飞)	PZ254V-11-03P_C2937625.1	LCSC	
27	1	NRST_ISO - FIT FOR STLINK / OPEN FOR ESP USB	JP8	HDR-TH_2P-P2.54-V-M		PZ254V-11-02P	XFCN(兴飞)	C492401	LCSC	1x2P 间距:2.54mm 方针 直插
28	1	VCP_TX_ISO - FIT FOR STLINK / OPEN FOR ESP USB	JP9	HDR-TH_2P-P2.54-V-M		PZ254V-11-02P	XFCN(兴飞)	C492401	LCSC	1x2P 间距:2.54mm 方针 直插
29	1	VCP_RX_ISO - FIT FOR STLINK / OPEN FOR ESP USB	JP10	HDR-TH_2P-P2.54-V-M		PZ254V-11-02P	XFCN(兴飞)	C492401	LCSC	1x2P 间距:2.54mm 方针 直插
30	1	T_VCC_ISO - FIT FOR STLINK / OPEN FOR USB-ONLY	JP11	HDR-TH_2P-P2.54-V-M		PZ254V-11-02P	XFCN(兴飞)	PZ254V-11-02P.1	LCSC	
31	1	4.7uH	L1	IND-SMD_L4.0-W4.0	4.7uH	SWPA4030S4R7MT	Sunlord(顺络)	SWPA4030S4R7MT.1	LCSC	
32	1	DEV_USER_LED_GREEN	LED1	LED0805-R-RD		KT-0805G	KENTO	C2297	LCSC	

33	1	STATION_STATUS_GREEN	LED2	LED0805-R-RD		KT-0805G	KENTO	C2297	LCSC	0805 翠绿 高亮
34	1	USB VBUS GREEN	LED3	LED0805-R-RD		KT-0805G	KENTO	0805G (Green).1	LCSC	
35	1	USB MODE GREEN	LED4	LED0805-R-RD		KT-0805G	KENTO	0805G (Green).1	LCSC	
36	1	AO3401A	Q1	SOT-23_L2.9-W1.3-P1.90-LS2.4-BR		AO3401A	AOS	C15127	LCSC	1个P沟道 耐压:30V 电流:4A
37	2	10Ω	R3,R4	R0603	10Ω	0603WAF100JT5E	UNI-ROYAL(厚声)	C22859	LCSC	10Ω ±1% 100mW 厚膜电阻
38	2	4.7kΩ	R5,R6	R0603	4.7kΩ	0603WAF4701T5E	UNI-ROYAL(厚声)	C23162	LCSC	厚膜电阻 4.7kΩ ±1%
39	2	10kΩ	R7,R18	R0603	10kΩ	0603WAF1002T5E	UNI-ROYAL(厚声)	C25804	LCSC	厚膜电阻 10kΩ ±1% 100mW
40	2	0Ω	R8,R9	R0603	0Ω	0603WAF0000T5E	UNI-ROYAL(厚声)	C21189	LCSC	厚膜电阻 0Ω ±1%
41	4	0Ω	R12,R13,R16,R17	R0603	0Ω	0603WAF0000T5E	UNI-ROYAL(厚声)	0603WAF0000T5E.1	LCSC	
42	1	10kΩ	R14	R0402	10kΩ	0402WGF1002TCE	UNI-ROYAL(厚声)	C25744	LCSC	
43	1	1kΩ	R15	R0402	1kΩ	0402WGF1001TCE	UNI-ROYAL(厚声)	C11702	LCSC	
44	1	1kΩ	R19	R0603	1kΩ	0603WAF1001T5E	UNI-ROYAL(厚声)	C21190	LCSC	1kΩ ±1% 100mW 厚膜电阻
45	1	5.1k CC1 Rd	R40	R0603	5.1kΩ	0603WAF5101T5E	UNI-ROYAL(厚声)	0603WAF5101T5E.1	LCSC	
46	1	5.1k CC2 Rd	R41	R0603	5.1kΩ	0603WAF5101T5E	UNI-ROYAL(厚声)	0603WAF5101T5E.1	LCSC	
47	1	100k SEL DEFAULT SWD	R42	R0603	100kΩ	0603WAF1003T5E	UNI-ROYAL(厚声)	0603WAF1003T5E.1	LCSC	
48	1	100kΩ	R43	R0603	100kΩ	0603WAF1003T5E	UNI-ROYAL(厚声)	C25803	LCSC	厚膜电阻 100kΩ ±1%
49	1	2.2k USB VBUS LED	R44	R0603	2.2kΩ	0603WAF2201T5E	UNI-ROYAL(厚声)	0603WAF2201T5E.1	LCSC	
50	1	2.2k USB MODE LED	R45	R0603	2.2kΩ	0603WAF2201T5E	UNI-ROYAL(厚声)	0603WAF2201T5E.1	LCSC	
51	1	100mΩ	RS1	R2512	100mΩ	WSL2512R1000FEA	VISHAY(威世)	C844904	LCSC	100mΩ ±1% 1W
52	1	DEV_RESET	SW1	SW-SMD_4P-L6.4-W6.3-P4.00-LS9.5		TS6635PA 250gf tactile switch	SHOU HAN(首韓)	C55213946	LCSC	6*6/6x6 防水 軽触スイッチ
53	1	DEV_USER	SW2	SW-SMD_4P-L6.4-W6.3-P4.00-LS9.5		TS6635PA 250gf tactile switch	SHOU HAN(首韓)	C55213946	LCSC	

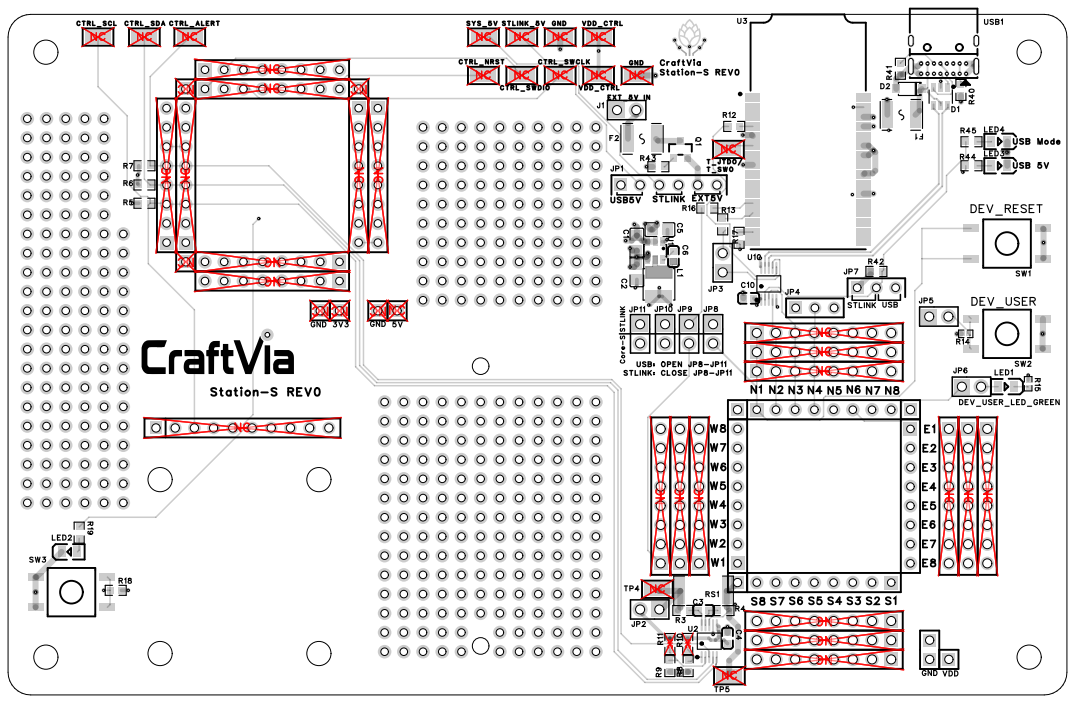
54	1	STATION_USER	SW3	SW-SMD_4P-L6.4-W6.3-P4.00-LS9.5		TS6635PA 250gf tactile switch	SHOU HAN(首韩)	C55213946	LCSC	6*6/6x6 防水轻触开关
55	1	AP63203WU-7	U1	TSOT-26_L2.9-W1.6-P0.95-LS2.8-BL		AP63203WU-7	DIODES(美台)	AP63203WU-7.1	LCSC	
56	1	INA226AIDGSR	U2	MSOP-10_L3.0-W3.0-P0.50-LS5.0-BL		INA226AIDGSR	TI(德州仪器)	C49851	LCSC	高侧/低侧测量双向电流/功率监视器，具有I²C兼容接口
57	1	STLINK-V3MODS	U3	COMM-SMD_STLINK-V3MODS_1		STLINK-V3MODS	ST(意法半导体)	STLINK-V3MODS.1	LCSC	
58	1	USB / SWD MUX	U10	MSOP-10_L3.0-W3.0-P0.50-LS5.0-BL		FSUSB42MUX	onsemi(安森美)	FSUSB42MUX.1	LCSC	
59	1	ESP32 USB-C / USB4085-GF-A (TH)	USB1	USC-C-SMD_USB4085-GF-A		USB4085-GF-A	GCT	C7095263	LCSC	Type-C 母弯插

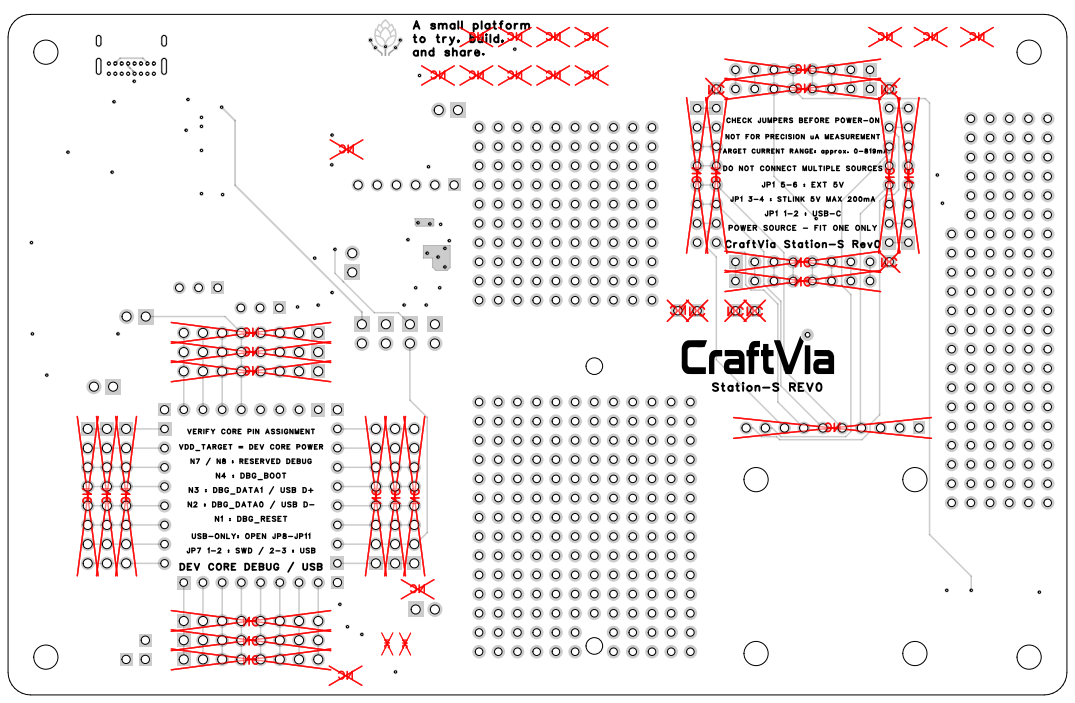
Description	Mounting Type
USB-C VBUS resettable PTC, 1.1A hold / 2.2A trip, 16V, 1812	
Voltage - Max:15V Current - Max:100A Hold Current:500mA Trip Current:1A Power Dissipation:800mW Resistance - Initial (Ri) (Min):150mΩ Resistance - Post Trip (R1) (Max):1Ω Time to Trip(Max):150ms Operating Temperature:-40℃~+85℃ Operating Temperature:-40℃~+	
Core-S DEV/DUT West socket W1-W8	
Core-S DEV/DUT East socket E1-E8; GPIO breakout	
Core-S DEV/DUT South socket S1-S8; GPIO breakout	
Core-S K-NW corner socket placeholder; final female socket TBD	
Core-S K-NE corner socket placeholder; final female socket TBD	
Core-S K-SW corner socket placeholder; final female socket TBD	
	Through Hole

Pin Structure:1x2P Pitch:2.54mm Row Spacing:- Circluar Pins/Square Pins:Square pin Mounting Type:Through Hole Number of Rows:1 Number of Pins:2P Current Rating:3A Voltage Rating (Max):250V Length of Mating Pin:6.1mm Insulation Height:2.5mm Length of End C	Through Hole
Pin Structure:1x2P Pitch:2.54mm Row Spacing:- Circluar Pins/Square Pins:Square pin Mounting Type:Through Hole Number of Rows:1 Number of Pins:2P Current Rating:3A Voltage Rating (Max):250V Length of Mating Pin:6.1mm Insulation Height:2.5mm Length of End C	Through Hole
Pin Structure:1x2P Pitch:2.54mm Row Spacing:- Circluar Pins/Square Pins:Square pin Mounting Type:Through Hole Number of Rows:1 Number of Pins:2P Current Rating:3A Voltage Rating (Max):250V Length of Mating Pin:6.1mm Insulation Height:2.5mm Length of End C	Through Hole
DEV/DUT USER LED; active high from N6/PA12	
Lamp Holder Type:2mm x 1.25mm square LED Illumination Color:Emerald Green Voltage - Forward(Vf):2.6V~3.1V Voltage - Forward(Vf):2.6V~3.1V Power(Watts):100mW Forward Current:5mA Wavelength - Dominant:513nm~528nm Wavelength - Dominant:513nm~528nm Peak Wavel	
Number:1 P-Channel Drain to Source Voltage:30V Current - Continuous Drain(Id):4A RDS(on):47mΩ@10V；85mΩ@2.5V；60mΩ@4.5V Pd - Power Dissipation:- Gate Threshold Voltage (Vgs(th)):900mV Gate Charge(Qg):14nC@10V Input Capacitance(Ciss):645pF Reverse Transfer C	
Type:Thick Film Resistor Resistance:10kΩ Tolerance:±1% Tolerance:±1% Power(Watts):100mW Voltage-Supply(Max):75V Temperature Coefficient:±100ppm/℃ Temperature Coefficient:±100ppm/℃ Operating Temperature:-55℃~+155℃ Operating Temperature:-55℃~+155℃	

SWCLK series configuration resistor, 0 ohm,SWDIO series configuration resistor, 0 ohm,VCP UART RX isolation / configuration resistor,VCP UART TX isolation / configuration resistor	
DEV_USER_BTN external pull-up to VDD_TARGET	
DEV_USER_LED current limiting resistor	
Type:Thick Film Resistor Resistance:1kΩ Tolerance:±1% Tolerance:±1% Power(Watts):100mW Voltage-Supply(Max):75V Temperature Coefficient:±100ppm/℃ Temperature Coefficient:±100ppm/℃ Operating Temperature:-55℃~+155℃ Operating Temperature:-55℃~+155℃	
Type:Thick Film Resistor Resistance:100kΩ Tolerance:±1% Tolerance:±1% Power\(Watts\):100mW Voltage-Supply(Max):75V Temperature Coefficient:±100ppm/℃ Temperature Coefficient:±100ppm/℃ Operating Temperature:-55℃~+155℃ Operating Temperature:-55℃~+155℃	
DEV/DUT reset pushbutton; momentary NO; shorts DEV_NRST to GND when pressed	Surface Mount,Vertical
DEV/DUT USER pushbutton; active low	
Circuit:SPST Actuator Style:Round Button Operating Force:2.5N Mounting Type:Surface Mount,Vertical Length:6.4mm Width:6.25mm Switch Height:3.5mm with Bracket:Without Bracket Contact Current:50mA Voltage Rating:12V Actuator/Cap Color:Blue With Lamp:No IP R	Surface Mount,Vertical

Connector Type:Type-C Protocol Standard:USB 2.0 Gender:Female Mounting Type:Right Angle Number of Ports:1 Current Rating-Power:5A Voltage Rating:48V Detection color:Black Operating Temperature:-40℃~+85℃ Operating Temperature:-40℃~+85℃	Right Angle


CTRL_SCL
CTRL_SDA
CTRL_ALERT
SYS_5V
STLINK_5V
GND
VDD_CTRL
CTRL_NRST
CTRL_SWCLK
CTRL_SWDIO
VDD_CTRL
GND
CraftVia
Station-S REVO
EXT_5V_IN
J1
F2
JP1
USB5V
STLINK_EXT5V
T_3V66
T_SWO
R12
R43
R16
R13
U3
USB1
D2
D1
R41
R40
R45
LED4
USB Mode
R44
LED3
USB 5V
DEV_RESET
SW1
U10
C10
JP4
JP7
R42
STLINK_USB
JP5
DEV_USER
SW2
R14
JP6
LED1
R15
DEV_USER_LED_GREEN
JP3
JP11
JP10
JP9
JP8
Core-STLINK
USB: OPEN JP8-JP11
STLINK: CLOSE JP8-JP11
N1 N2 N3 N4 N5 N6 N7 N8
W8 W7 W6 W5 W4 W3 W2 W1
E1 E2 E3 E4 E5 E6 E7 E8
S8 S7 S6 S5 S4 S3 S2 S1
TP4
JP2
RS1
C7
R4
R3
U2
TP5
GND
VDD
R7
R6
R5
GND_3V3
GND_5V
CraftVia
Station-S REVO
LED2
R19
SW3
R18


A small platform
to try, build,
and share.
CHECK JUMPERS BEFORE POWER-ON
NOT FOR PRECISION uA MEASUREMENT
TARGET CURRENT RANGE: approx. 0–819mA
DO NOT CONNECT MULTIPLE SOURCES
JP1 5–6 : EXT 5V
JP1 3–4 : STLINK 5V MAX 200mA
JP1 1–2 : USB-C
POWER SOURCE – FIT ONE ONLY
CraftVia Station-S RevO
CraftVia
Station-S REVO
VERIFY CORE PIN ASSIGNMENT
VDD_TARGET = DEV CORE POWER
N7 / N8 : RESERVED DEBUG
N4 : DBG_BOOT
N3 : DBG_DATA1 / USB D+
N2 : DBG_DATA0 / USB D–
N1 : DBG_RESET
USB-ONLY: OPEN JP8–JP11
JP7 1–2 : SWD / 2–3 : USB
DEV CORE DEBUG / USB

